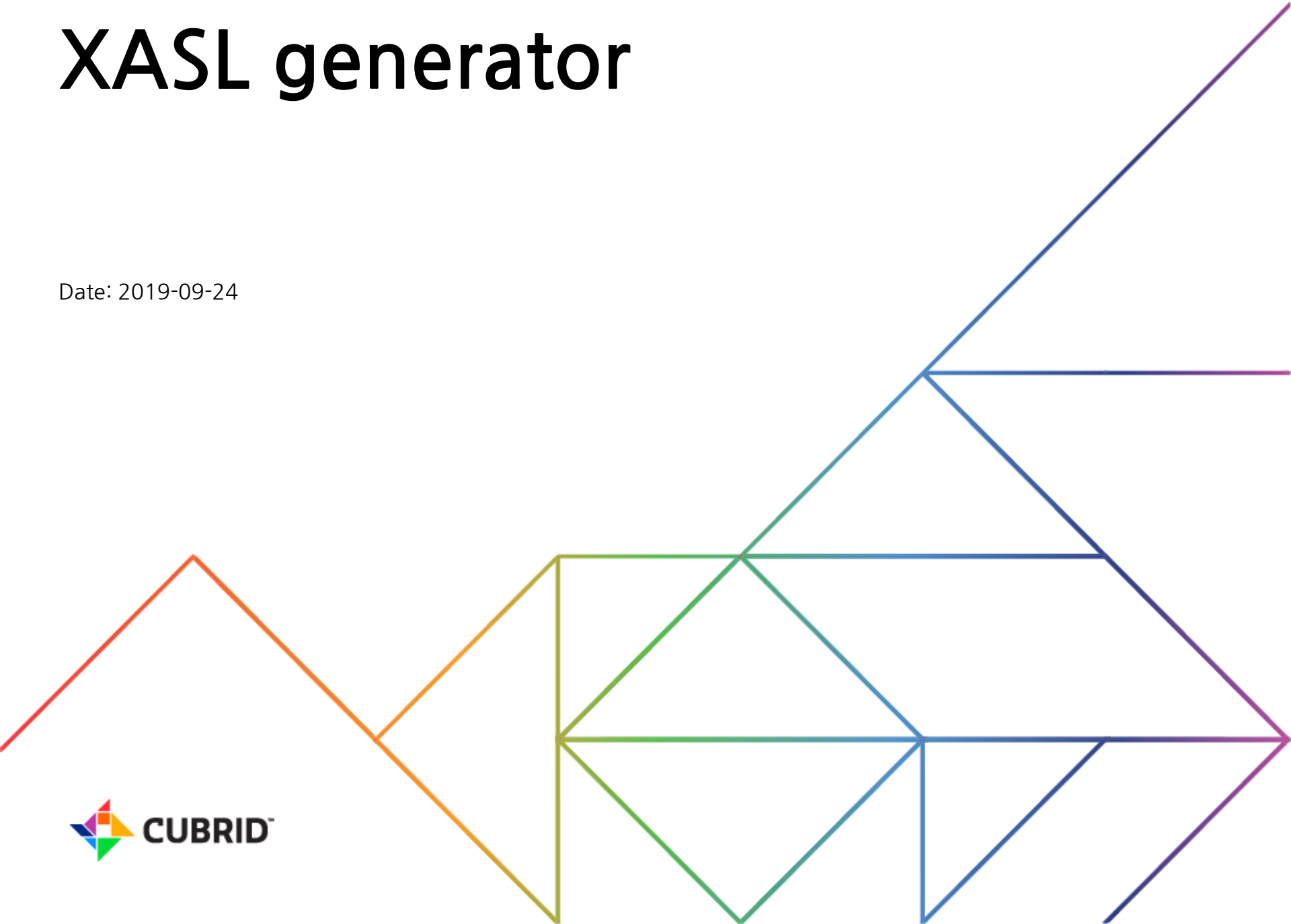
XASL generator

Date: 2019-09-24



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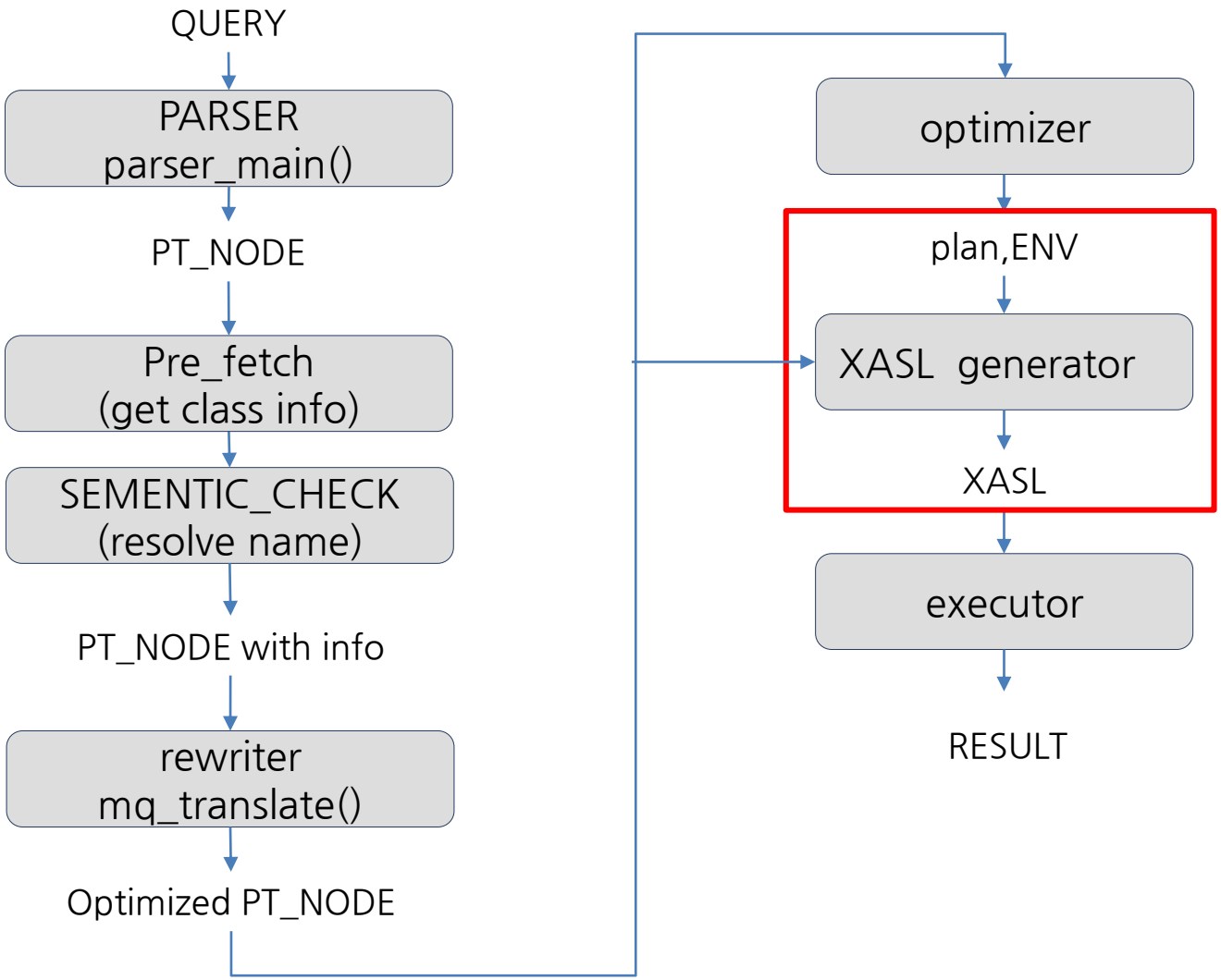
Query process



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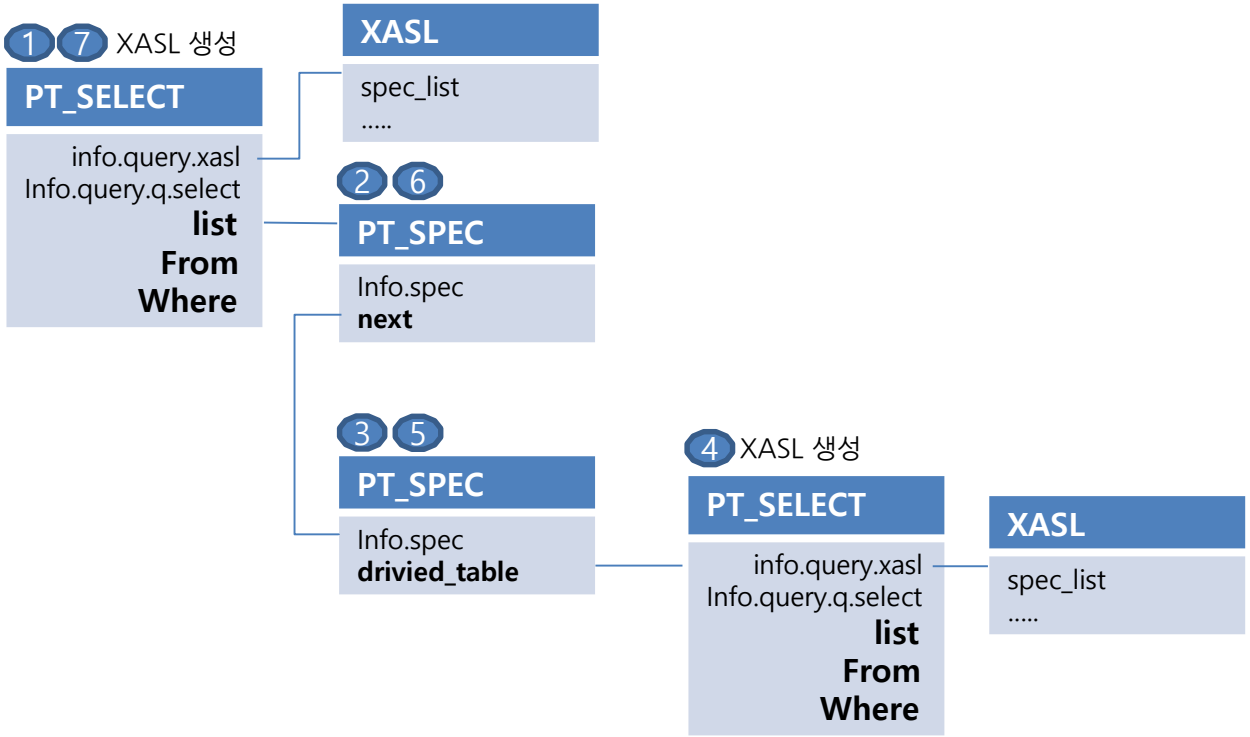
XASL 수행 방식

- parser_walk_tree() 함수의 post 함수로 수행하며 PT_SELECT type일 때 수행된다.
- Sub query의 경우 먼저 optimization 및 XASL generation 된다.

Query with subquery

```
SELECT 1
  FROM table a,
          (select 1 from tbl2 …) b
WHERE …
```

parser_walk_tree()



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XASL

- eXtensional Access Specification Language
- SQL : 데이터 가져오는 순서가 없음
- XASL : plan 정보를 사용하여 순서가 있는 절차적 프로시져

PROC TYPE
UNION_PROC
DIFFERENCE_PROC
INTERSECTION_PROC
OBJFETCH_PROC
BUILDLIST_PROC
BUILDVALUE_PROC
SCAN_PROC
MERGELIST_PROC
UPDATE_PROC
DELETE_PROC
INSERT_PROC
CONNECTBY_PROC
DO_PROC
MERGE_PROC
BUILD_SCHEMA_PROC
CTE_PROC

- BUILDLIST_PROC : ROW가 여러건이 될 수 있는 SQL
- BUILDVALUE_PROC : ROW가 한 건 조회되는 SQL
  ex) select sum(col1) from tab
- SCAN_PROC : TEMP 영역을 사용하지 않음. JOIN에 사용됨.

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XASL(aptr, dptr, scan_ptr)

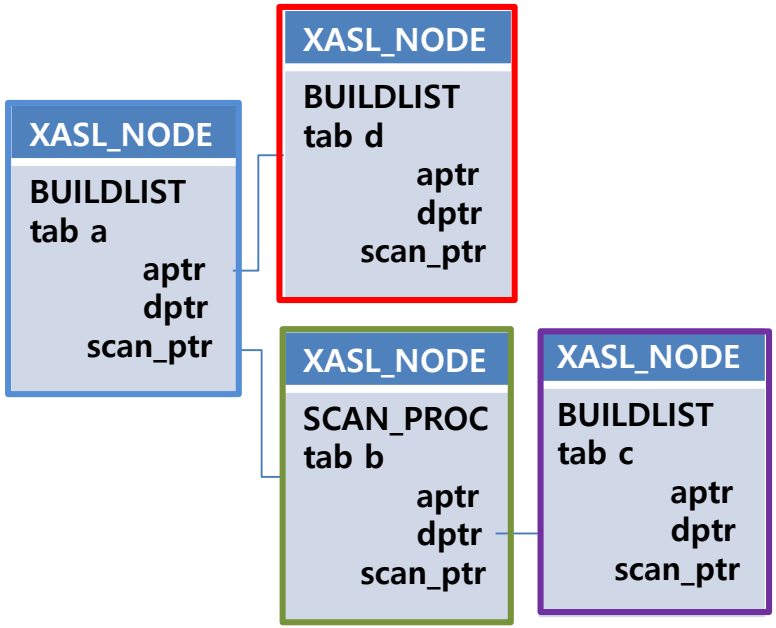
- aptr : uncorrelated subquery
- dptr : correlated subquery
- scan_ptr : join (outer scan_ptr → inner)

QUERY

```
SELECT
(select col1 from tab c
where col1 = b.col1)
FROM tab a, tab b
WHERE a.col2 = b.col2
   AND a.col1 =
(select col1 from tab d)
```

join 순서 : a → b

XASL



EXECUTE

SCAN aptr
SCAN node
 for each row
   SCAN dptr
   SCAN scan_ptr

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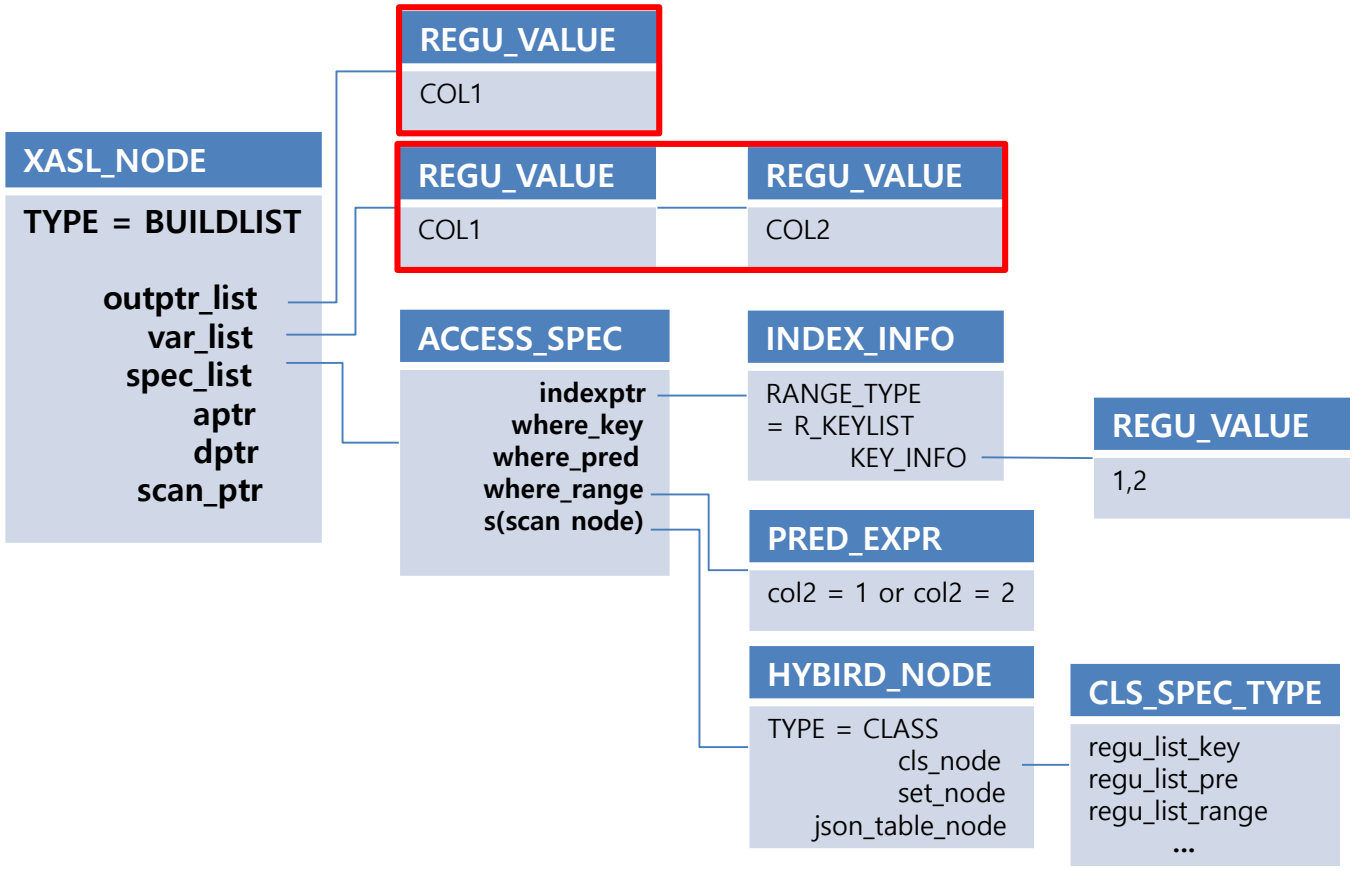
XASL node

- outptr_list : select list
- var_list : table's columns
- spec_list : access spec list (class oid, index info, predicate)

QUERY

SELECT col1
FROM table a
WHERE col2 in (1,2)

XASL



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REGULAR VARIABLE (pt_to_regu_variable())

- expression을 포함하여 모든 데이터 타입을 저장할 수 있는 변수
- EXECUTE 단계에서 데이터를 저장하거나 조회하는데 사용됨.
- fetch_peek_dbval() : regu → db_value, regular variable 에서 데이터 가져오는 함수

QUERY

```
SELECT col1, col2
  FROM table a
WHERE col1 in (1,2)
```

XASL

COL1
PT_NAME

REGU_VARIABLE

TYPE = CONSTANT
domain
UNION
dbval
dbvalptr
arithptr
funcp

DB_VALUE

domain.type = 1
DB_DATA
i (int)
d (double)
char
.....

COL1
PT_NAME

REGU_VARIABLE

TYPE = ATTR_ID
domain
UNION
attr_descr
id = 0
type = INT
cache_dbval

(1,2) 1:
PT_POS_VAL

REGU_VARIABLE

TYPE = POS_VALUE
domain
UNION
val_pos = 0

2:

REGU_VARIABLE

TYPE = POS_VALUE
domain
UNION
val_pos = 1

VAL_DESCR

DB_VALUE 1	DB_VALUE 2

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REGULAR VARIABLE (pt_to_regu_variable())

- PT_EXPR은 OPERATION에 따라 다양하게 변경됨.

QUERY	XASL

SELECT ‘park’
FROM tab a
WHERE col1 = nvl(col1,0)

‘park’
PT_VALUE

REGU_VARIABLE
TYPE = DBVAL
domain
UNION
dbval

DB_VALUE
domain.type = 25
DB_DATA
char = ‘park’

nvl(col1,0)
PT_EXPR

REGU_VARIABLE
TYPE = INARITH
domain
UNION
arithptr

ARITH_NODE
domain
op = nvl
value
leftptr
rightptr
thirdptr

REGU_VALUE
COL1

REGU_VALUE
0

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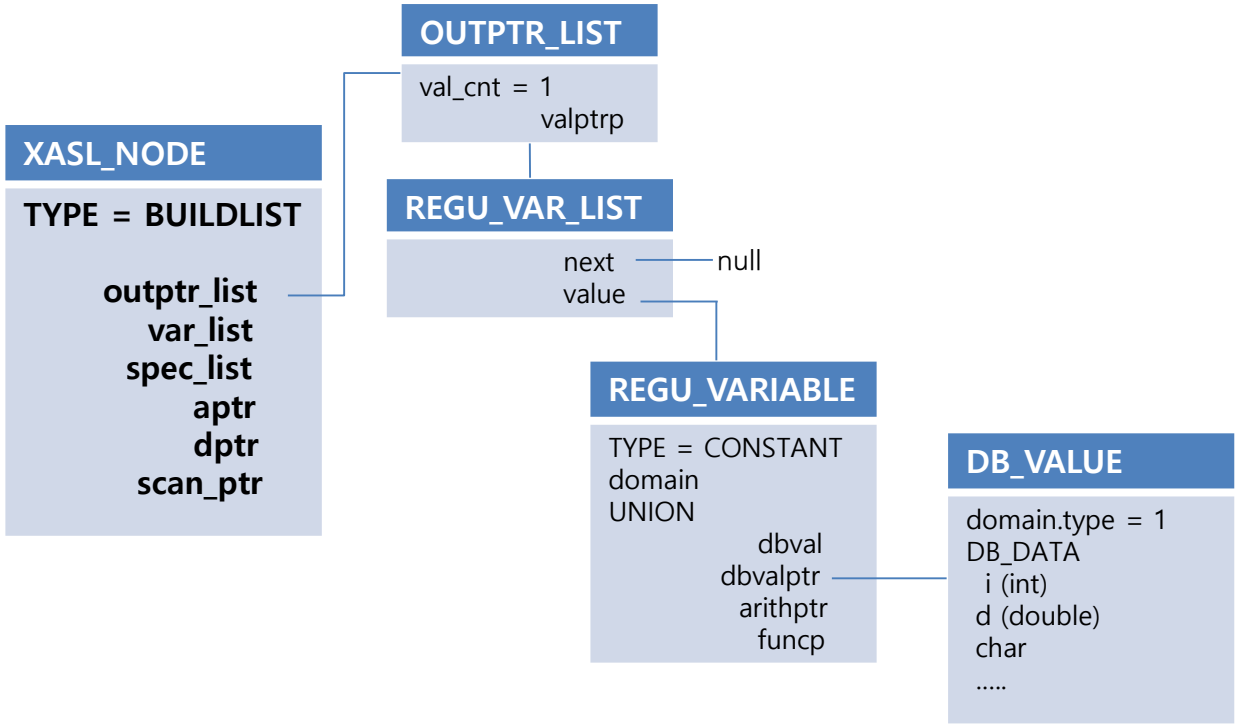
OUTPTR_LIST (pt_to_outlist())

- select list의 column, 상수

QUERY

SELECT col1
FROM table a
WHERE col2 in (1,2)

XASL



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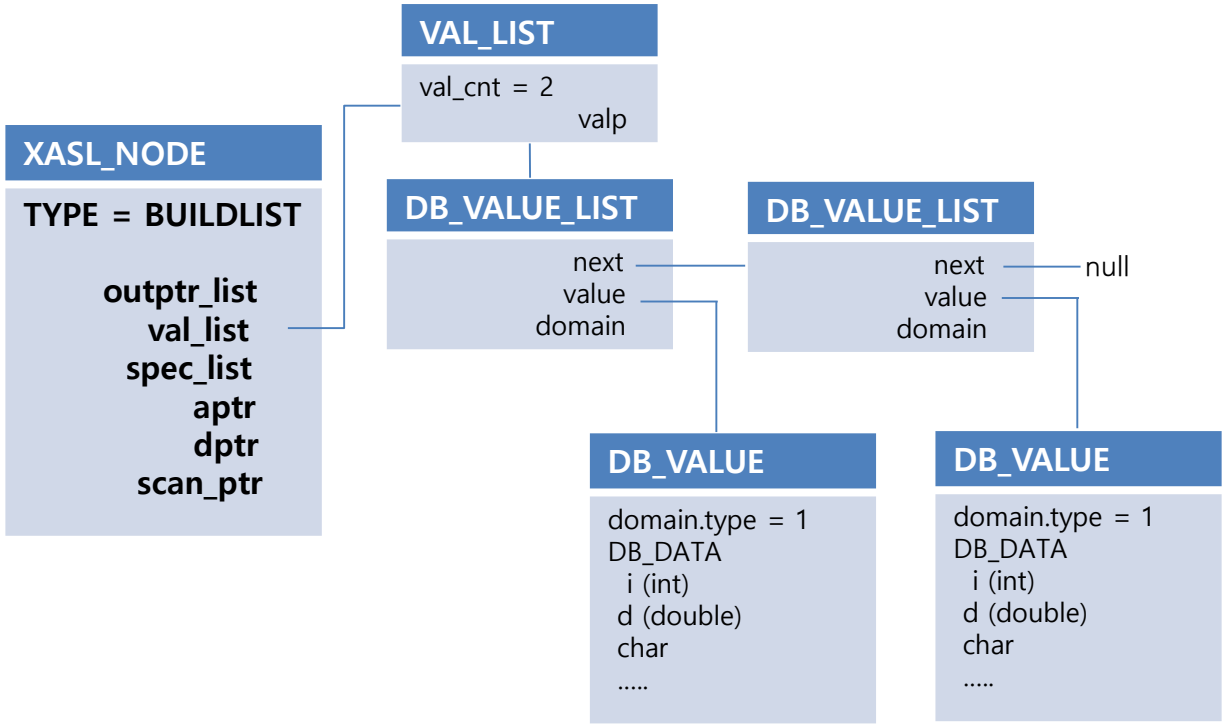
VAL_LIST (pt_to_val_list())

- 테이블의 column list
- outptr_list와 동일한 column은 DB_VALUE를 공유함.

QUERY

SELECT col1
FROM table a
WHERE col2 in (1,2)

XASL



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SPEC_LIST (add_access_spec ())

- indexptr : index scan시 필요한 정보 (btree id, range_type, key_info)
- where_range, where_key, where_pred : predicate (index key, index filter, heap)
- s(scan node) : 조회 대상 node 정보

QUERY

SELECT col1
FROM table a
WHERE col2 in (1,2)

XASL

XASL_NODE

TYPE = BUILDLIST

outptr_list
val_list
spec_list
aptr
dptr
scan_ptr

ACCESS_SPEC

type = CLASS
access = INDEX

indexptr
where_key
where_pred
where_range
s(scan node)

INDEX_INFO

RANGE_TYPE
= R_KEYLIST
KEY_INFO

REGU_VAR

1,2

PRED_EXPR

col2 = 1 or col2 = 2

HYBIRD_NODE

TYPE = CLASS
cls_node
set_node
json_table_node

CLS_SPEC_TYPE

regu_list_key
regu_list_pred
regu_list_range
...

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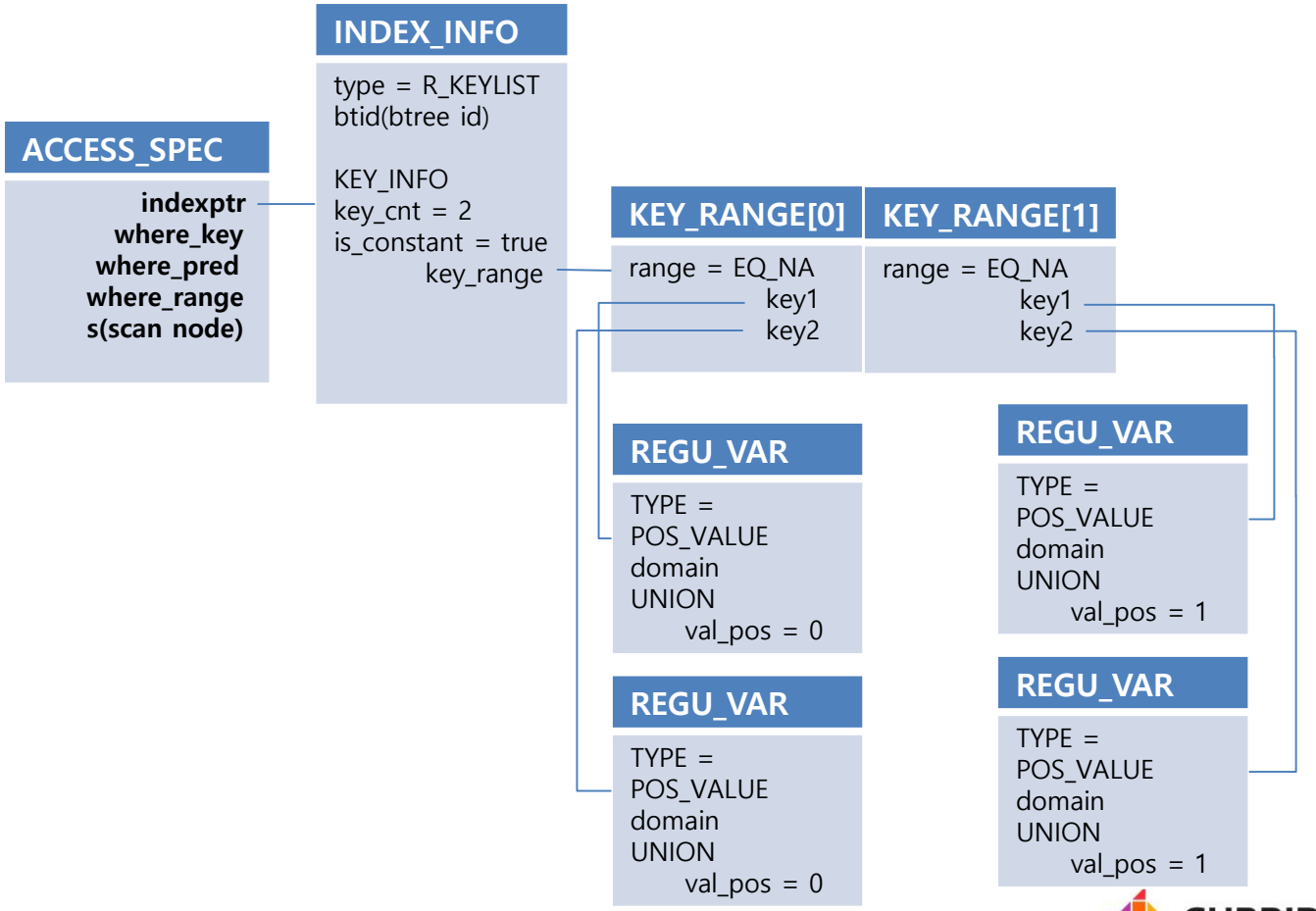
INDEXPTR (pt_to_index_info())

- RANGE_TYPE : R_KEY(=), R_RANGE(>), R_KEYLIST(IN), R_RANGE_LIST(> OR <)
- KEY1 : lower limit
- KEY2 : upper limit ex) BETWEEN 1 and 2 ➔ key1 = 1, key2 = 2

QUERY

SELECT col1
FROM table a
WHERE col2 in (1,2)

XASL



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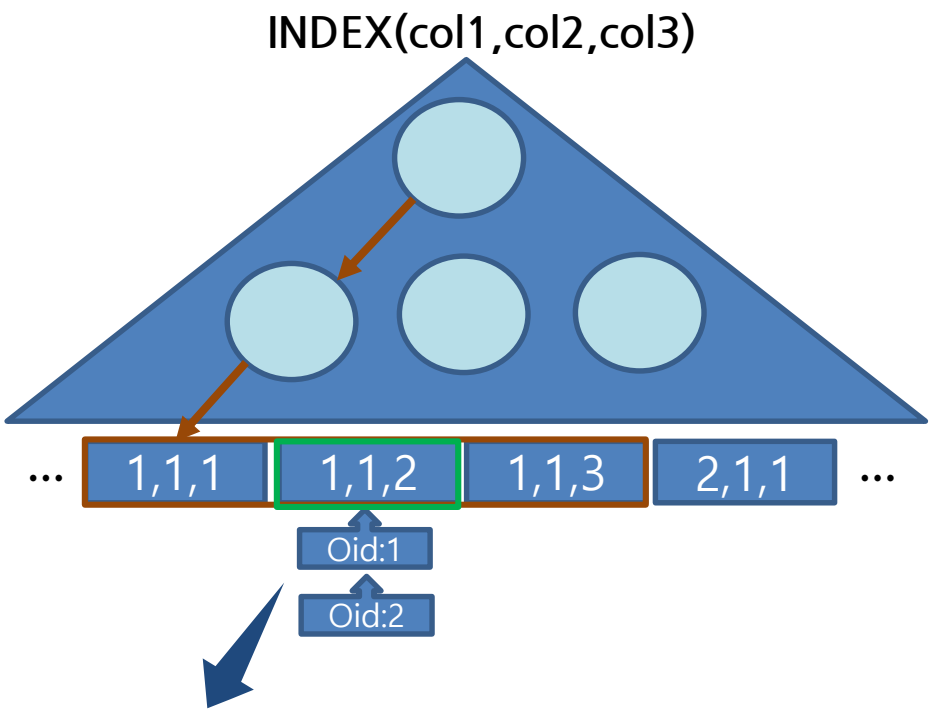
WHERE_RANGE, WHERE_KEY, WHERE_PRED

- WHERE_RANGE : INDEX 수직 스캔가능한 PREDCATE
- WHERE_KEY : INDEX 수평 스캔 가능한 PREDCATE
- WHERE_PRED : 데이터 영역에서 비교가 필요한 PREDCATE

```
SELECT * FROM tbl
WHERE col1 = 1      ➔RANGE
    AND col3 = 2    ➔KEY
    AND col4 = 2    ➔PRED
```

유형	테이블	인덱스	검색 조건
질의			
Index Scan	tbl	cub1	
index 2			tbl.col1=1
sargs			tbl.col4=2
filter			tbl.col3=2



DATA

Oid	Col1	Col2	Col3	Col4
1	1	1	2	5
2	1	1	2	2

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WHERE_RANGE, WHERE_KEY, WHERE_PRED

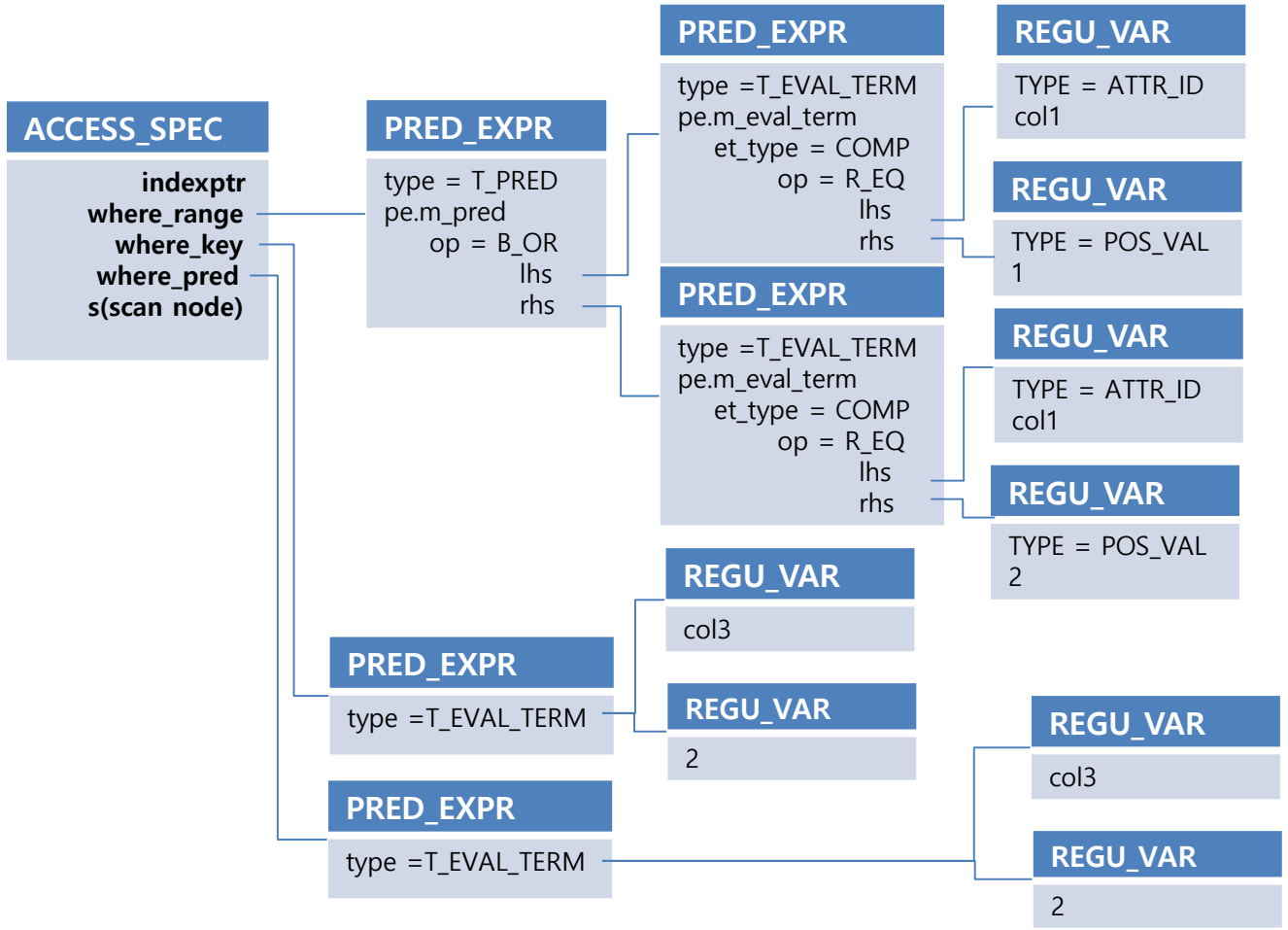
- PARSER에서 생성된 PREDICATE PT_NODE를 PRED_EXPR 구조체로 변경한다.
- PT_NODE보다 PRED_EXPR이 가벼운 구조체

QUERY

```
SELECT 1
  FROM table a
WHERE col1 in (1,2)   ➔RANGE
    AND col3 = 2      ➔KEY
    AND col4 = 2      ➔PRED

index(col1,col2,col3)
```

XASL



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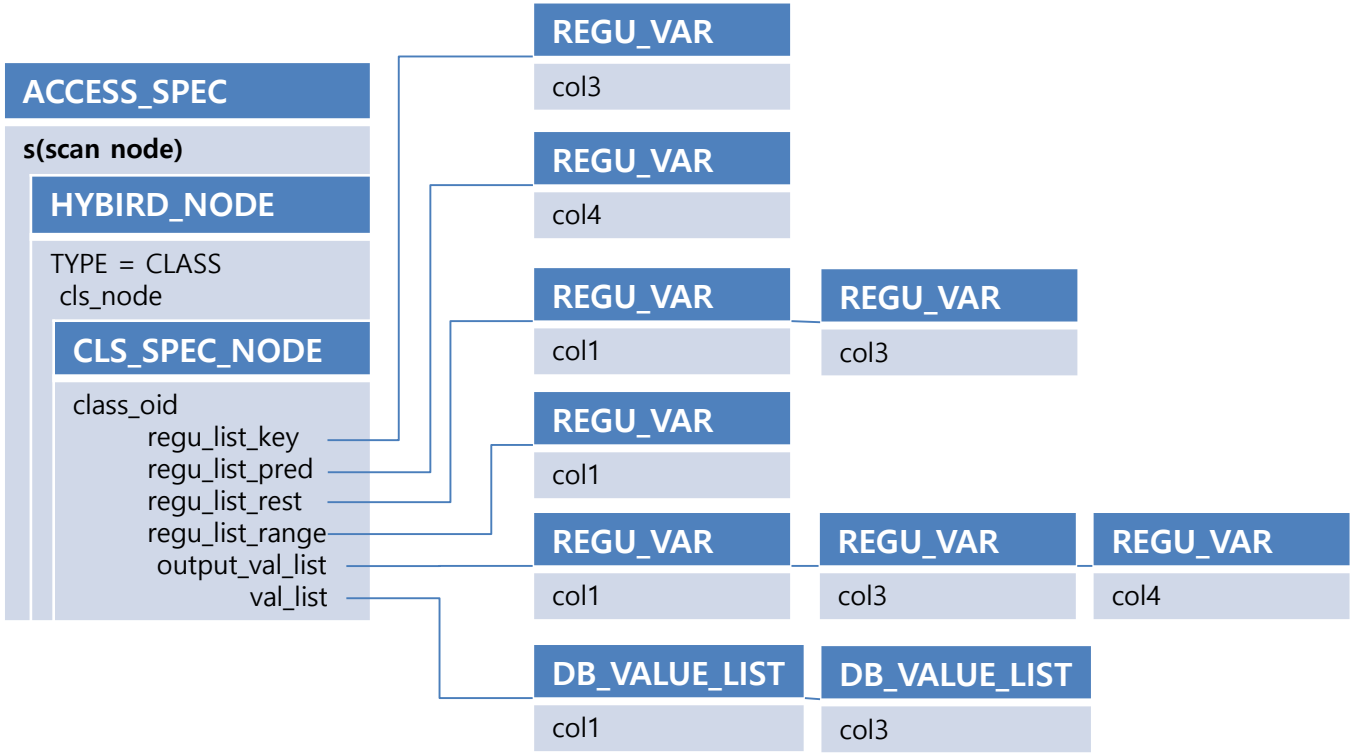
S(SPEC NODE)

- TYPE에 따라 CLASS(table) node, SET node, JSON node등으로 구분됨.
- 동일 컬럼의 regu_var일 경우 db_value의 값을 공유함.

QUERY

SELECT 1
FROM table a
WHERE col1 in (1,2) ➔RANGE
AND col3 = 2 ➔KEY
AND col4 = 2 ➔PRED

XASL



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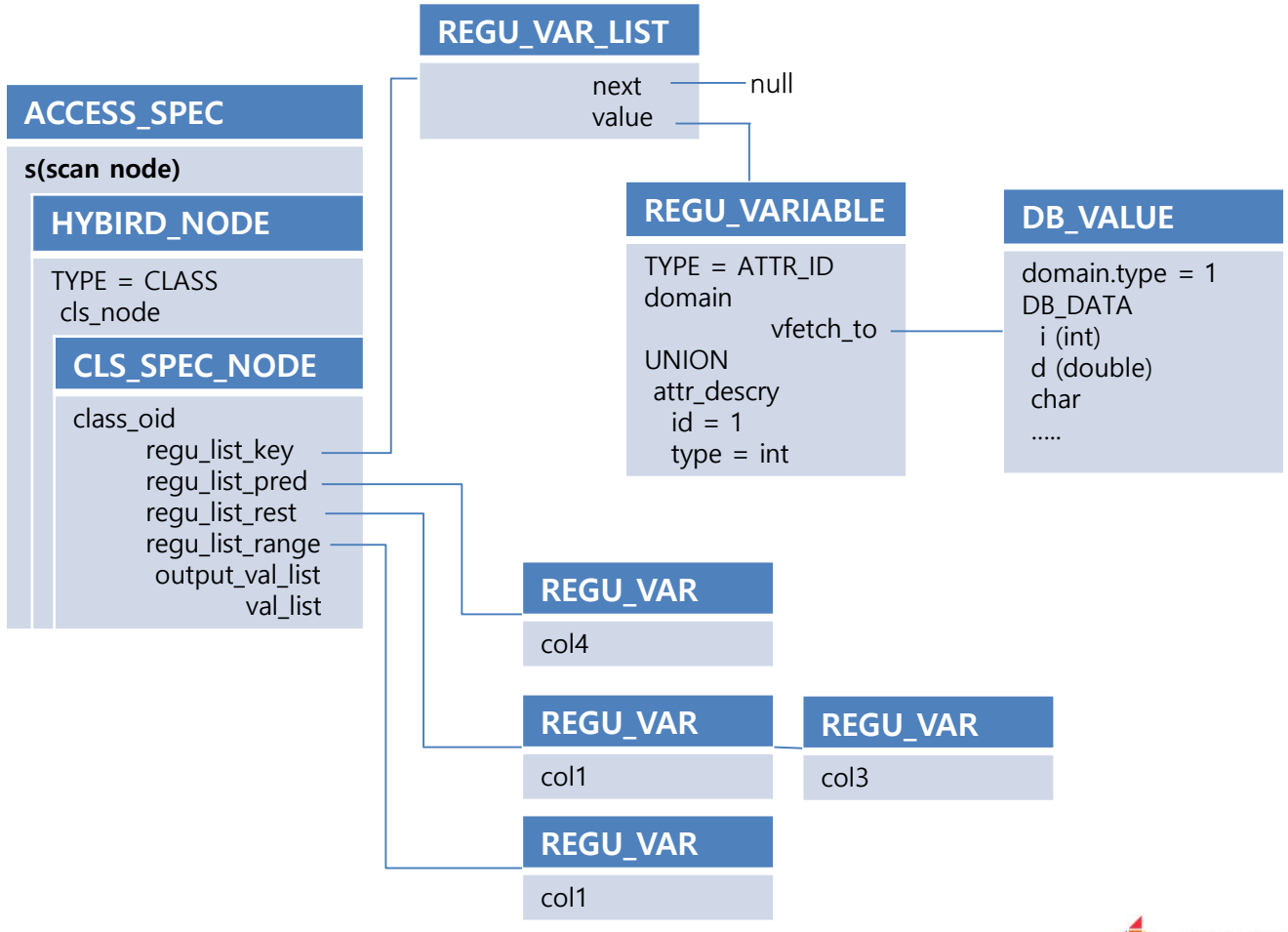
S(SPEC NODE) regu_list

- range,key,pred : 인덱스 수직,수평,데이터 영역 스캔관련 컬럼
- rest : range + key

QUERY

SELECT 1
FROM table a
WHERE col1 in (1,2) ➔RANGE
AND col3 = 2 ➔KEY
AND col4 = 2 ➔PRED

XASL



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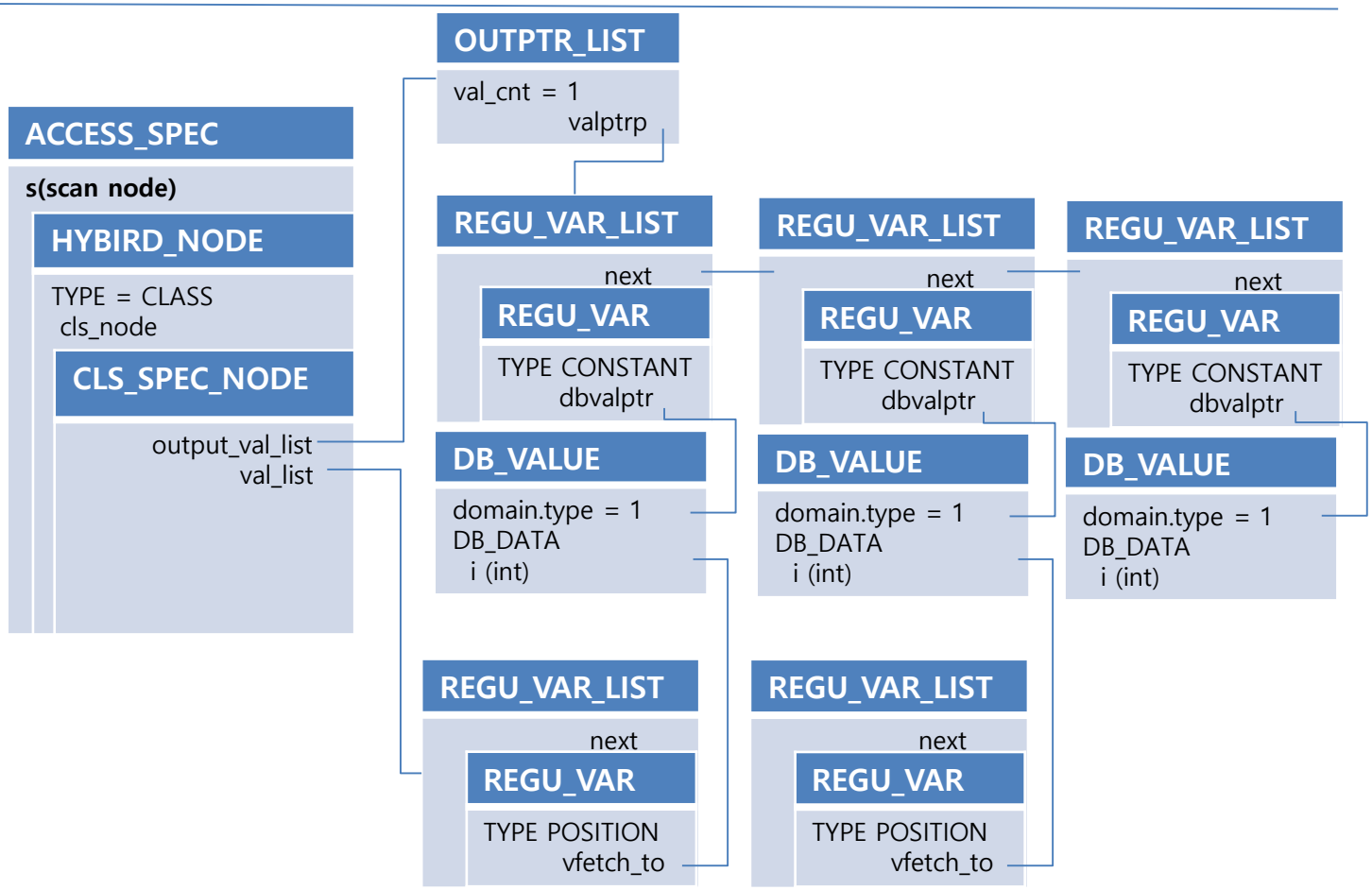
S(SPEC NODE) output_val_list, val_list

- output_val_list : 해당 spec의 컬럼 list, writer
- val_list : rest에 해당하는 컬럼 list, reader

QUERY

```
SELECT 1
  FROM table a
WHERE col1 in (1,2)
    AND col3 = 2
    AND col4 = 2
```

XASL



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SINGLE TABLE EXECUTE

- SINGLE TABLE에서 INDEX SCAN 예시.
- EXECUTOR에 대한 분석 필요.

XASL

ACCESS_SPEC

indexptr
where_key
where_pred
where_range
s(scan node)

INDEX_INFO

RANGE_TYPE
= R_KEYLIST
KEY_INFO

REGU_VAL

1,2

PRED_EXPR

col3=3

PRED_EXPR

col2 = 1 or col2 = 2

CLS_SPEC

regu_list_key
regu_list_pred
regu_list_rest
regu_list_range
output_val_list
val_list

REGU_VAR

col3

REGU_VAR

col2

REGU_VAR

col2

REGU_VAR

col3

EXECUTOR

for index scan using KEY_INFO(1,2)
  fetch row from OID
  for each row
    evaluation(PRED_EXPR(COL3=3))

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JOIN

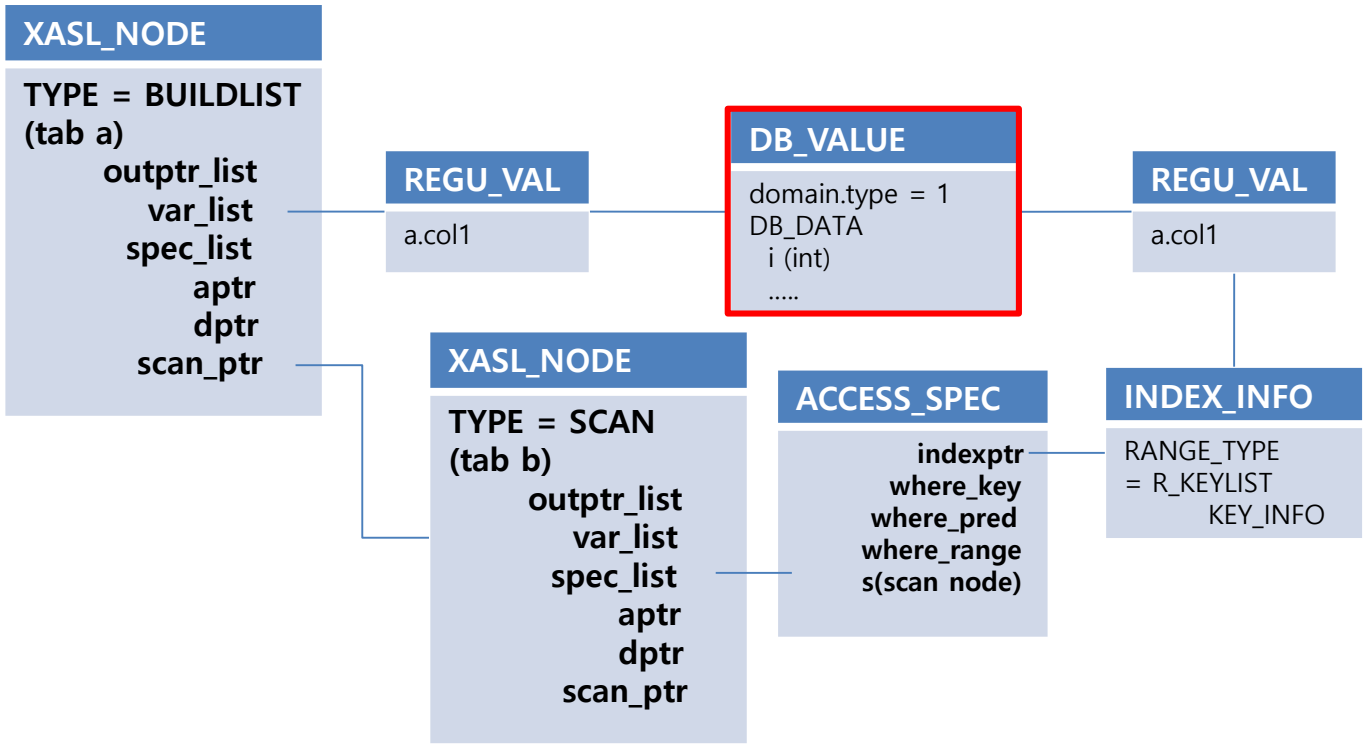
- join 순서에 따라 scan_ptr에 XASL 정보 저장됨
- BUILDLIST_PROC : temp 저장공간에 결과값 저장
- SCAN_PROC : temp 저장 공간 사용하지 않음

QUERY

SELECT col1
FROM tab a, tab b
WHERE a.col1 = b.col1

plan
outer: sscan
node a
inner : iscan
node b

XASL



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UNCORRELATED SUBQUERY

- aptr 포인터에 uncorrelated subquery xasl 저장됨.
- subquery는 BUILDLIST_PROC으로 생성됨.

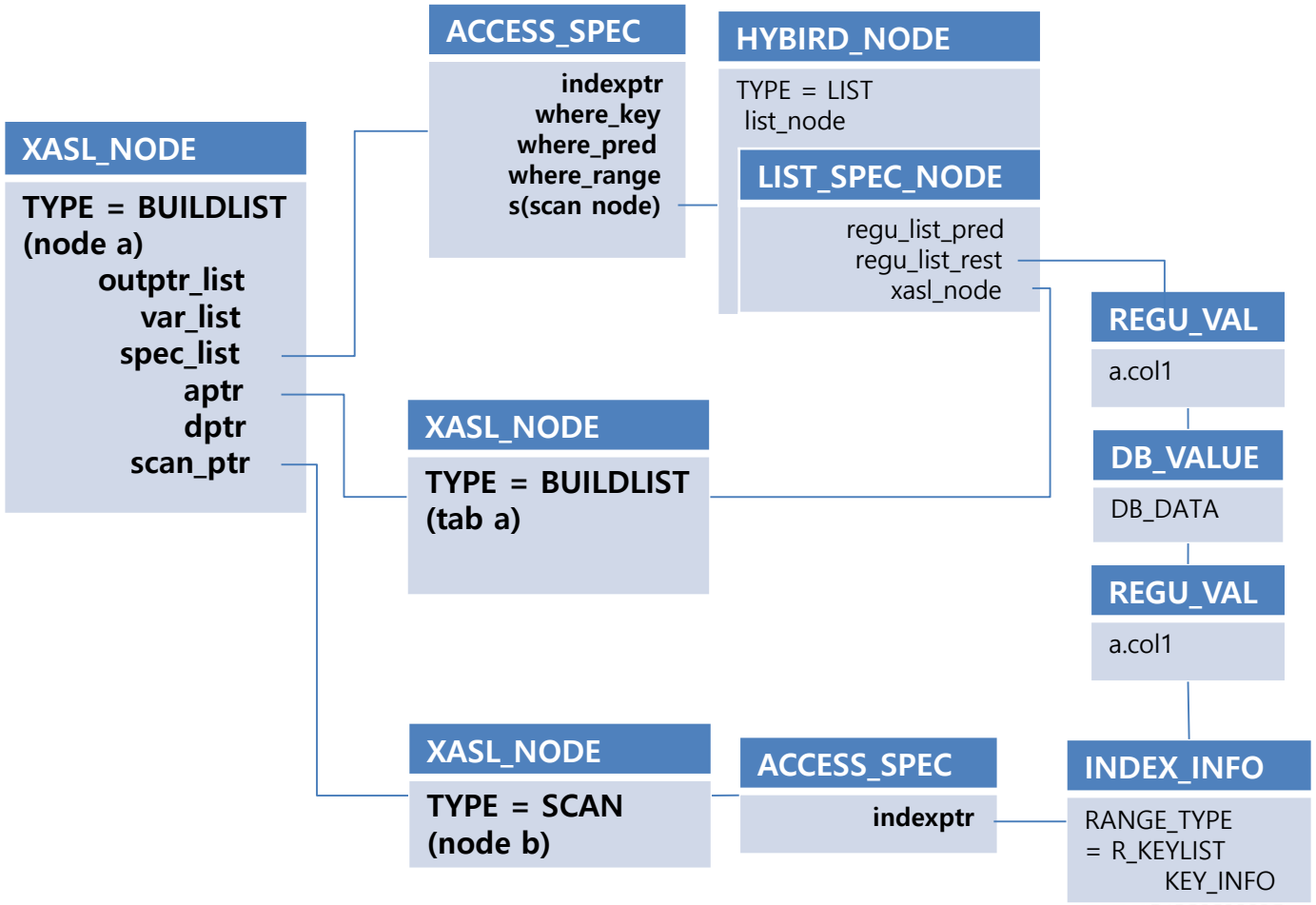
QUERY

```
SELECT a.col1
 FROM
 (select col1 from tab a
  where col2 =3) a
  ,tab b
WHERE a.col1 = b.col1
```

plan
sscan tab a

plan
outer: sscan
node a
inner : iscan
node b

XASL



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CORRELATED SUBQUERY

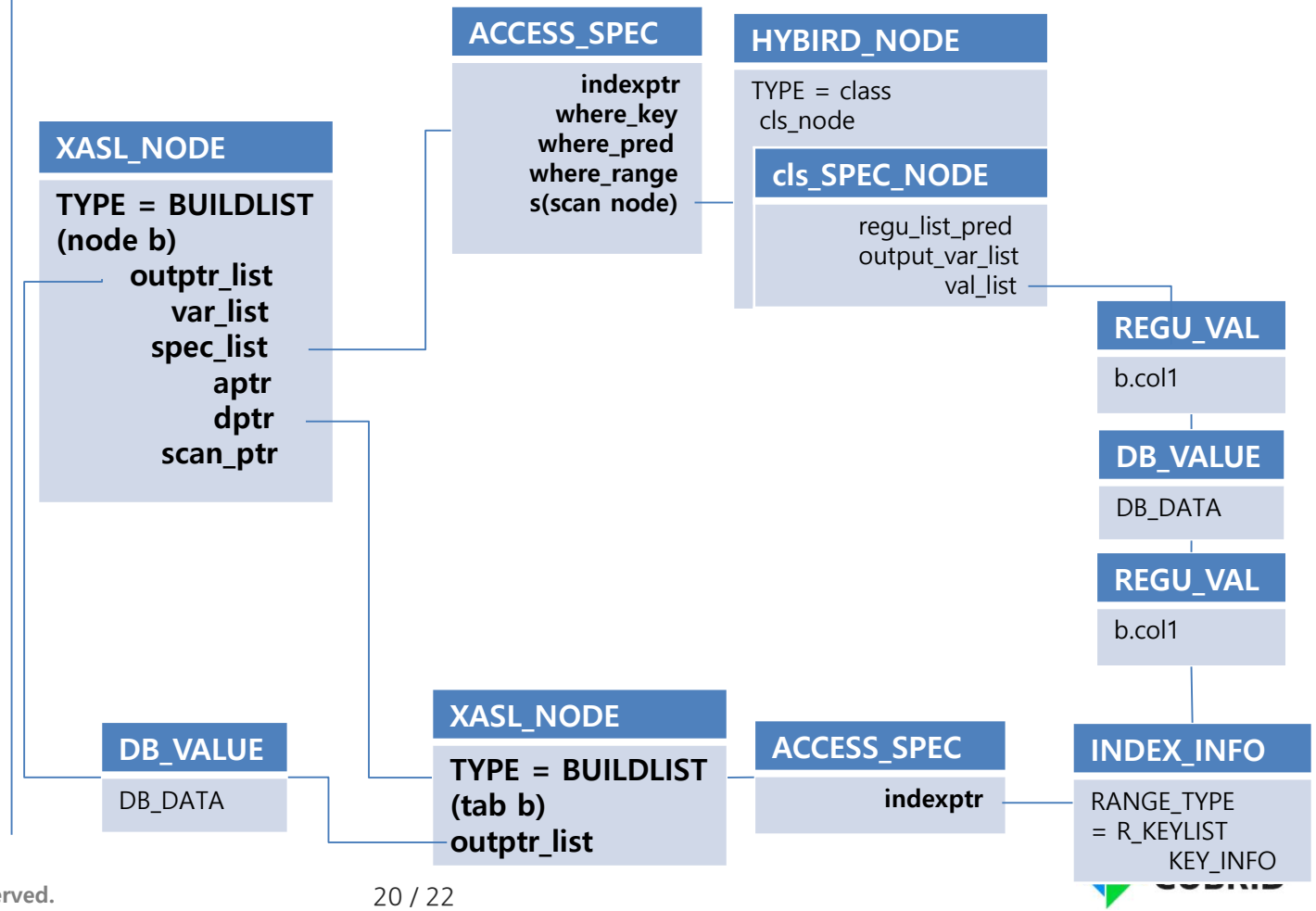
- dptr 포인터에 correlated subquery xasl 저장됨.
- MAIN proc 진행후 dptr proc 진행함

QUERY

SELECT
(select col1 from tab a
  where col1 = b.col1)
FROM tab b

plan
iscan tab a

plan
sscan node b



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XASL generator 진행 절차

```
pt_to_buildlist_proc
  pt_to_outlist(select.list)                  <-- outlist 생성
  pt_set_aptr(select)                         <-- aptr 생성
  pt_gen_optimized_plan(select,plan)  <-- 나머지 생성
   qo_to_xasl(plan)
    gen_outer(env,plan)
    pt_set_dptr(select.list)

gen_outer(inner_scans)
  case QO_PLANTYPE_SCAN
    add_access_spec(env,plan)   <-- spec list 생성
    add_scan_proc(inner_scans) <-- scan_ptr 생성
    add_subqueries                    <-- aptr, dptr 생성

  case QO_PLANTYPE_JOIN:
    outer_plan = plan->plan_un.join.outer;
    inner_plan = plan->plan_un.join.inner;

    inner_scan = gen_inner(inner_plan)
    gen_outer(inner_scan)
```

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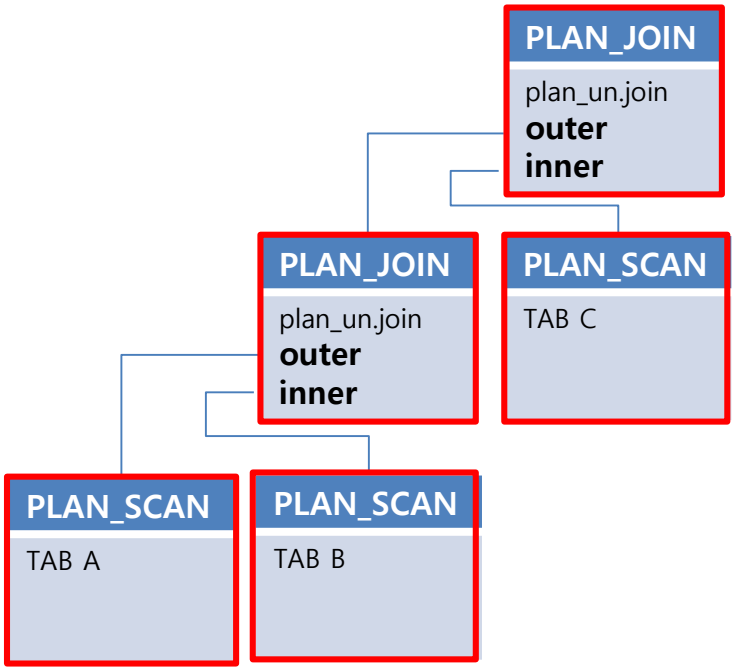
PLAN to XASL

- get_outer 함수가 재귀 호출하며 XASL 생성함.

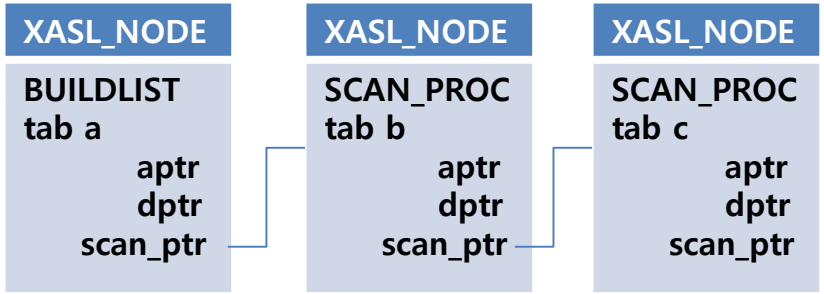
PLAN

select 1 from A,B,C where …
join order : A->B->C



XASL

```
gen_outer(inner_scan)
  case PLAN_SCAN
    scan_ptr = inner_scan
    spec 생성
  case PLAN_JOIN:
    inner = gen_inner(inner_scan)
    gen_outer(inner)
```



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